

Quality Management

1 — Introduction and the European framework

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BSc in Engineering Management • Università degli Studi di Perugia

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What this course is about

Five blocks, in this order

1. Regulations and the EU framework
2. History of quality management
3. Accreditation, certification and reliability
4. Statistical tools for quality management
5. Measurement techniques for quality control

The question behind all five

How do you convince someone who has never met you that your product does what you say it does?

Who is teaching

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- PhD in Mechanical Measurement for Engineering and Space
- Postdoctoral Research Associate, University of Illinois (NASA-funded)
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ASK QUESTIONS

During the lecture, not after it. If something does not make sense, it is far more likely that I explained it badly than that you missed it.



Who else you will hear from

Three guest seminars — 15 October, 5 November, 26 November

Quality management is not an academic subject. It is what a company does on a Tuesday because a customer, a regulator or an auditor requires it.

Three sessions are given by people who do this for a living — in manufacturing, in aerospace supply, and in the national metrology system.

These count

What the guests present is part of the course and is examinable. They are also the clearest view you will get of how the rest of this material is actually used.

Speakers are confirmed a few weeks ahead and announced on the course page.

How the course works

When

- Monday 10:30–13:30, Room 3
- Thursday 11:30–13:30, Room 2
- 21 September – 3 December

Prerequisites

None.

Exam

Oral examination.

Material

- These slides, published after each lecture
- Your own notes
- *Juran's Quality Control Handbook* — reference, not required reading cover to cover

Everything is online

github.com/LolloCappo/student-hub — schedule, slides, and the notebooks for the hands-on session.

The words we will use

Used loosely in industry, precisely in this course

- **Quality** — conformance to requirements. Not “good”, not “expensive”: *conforming to what was specified.*
- **Quality Management** — the activities by which an organisation sets its quality policy and carries it out.
- **Quality Control (QC)** — checking that the output conforms. A *corrective* tool, applied to the product: inspection, testing, measurement.
- **Quality Assurance (QA)** — building confidence that requirements *will* be met. A *preventive* tool, applied to the process: audits, procedures, checklists.

THE DISTINCTION THAT MATTERS

QC finds the defective bearing. QA is the reason the line stopped producing defective bearings. Confusing the two is the most common mistake at the exam.

The words we will use

continued

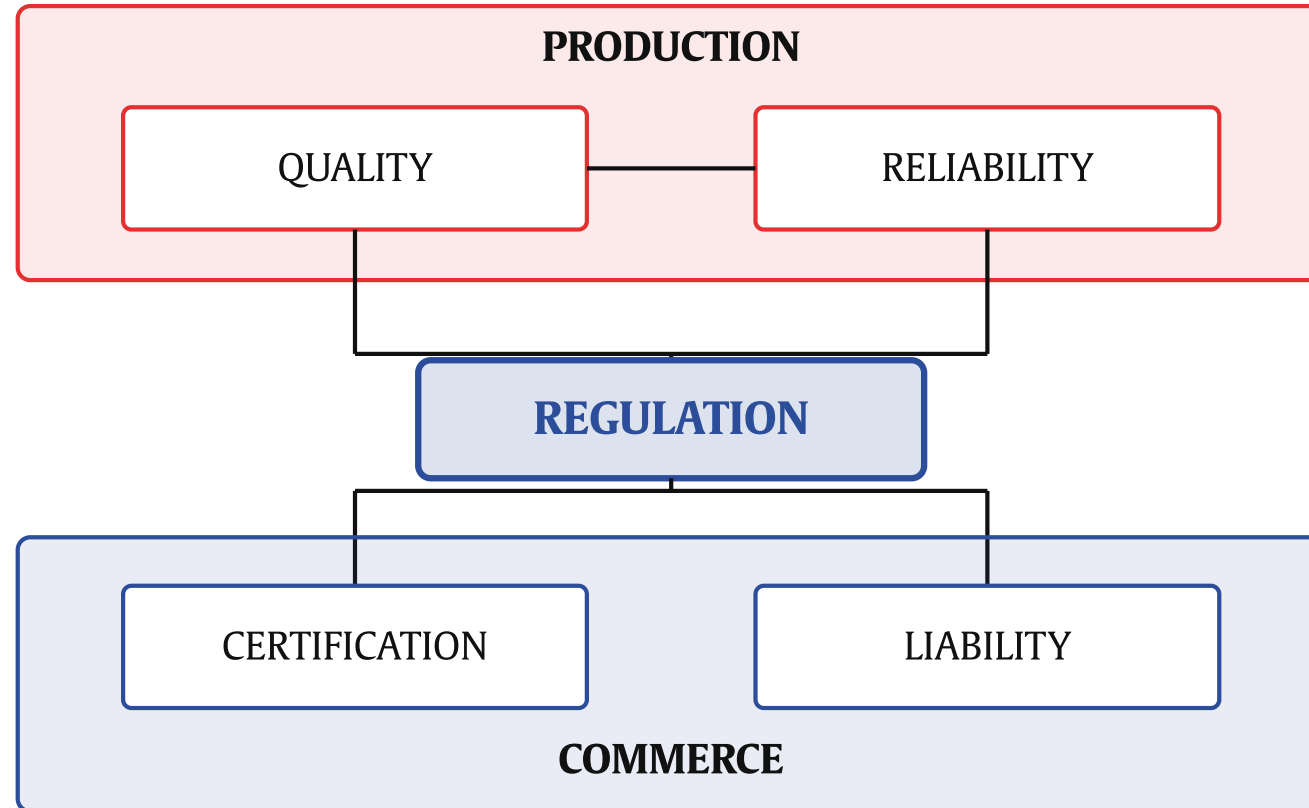
- **Quality Management System (QMS)** — the whole structure: policies, objectives, planning, improvement, with both QA and QC inside it. This is what ISO 9001 certifies.
- **Process** — any activity that uses resources to turn inputs into outputs. Quality management acts on processes, not on people.
- **Reliability** — the ability to perform the required function, under stated conditions, *for a stated period of time*.

QUALITY IS NOT RELIABILITY

A bolt that meets every dimension on the drawing the day it ships has **quality**. A bolt still holding after ten years of vibration and salt spray has **reliability**. Different properties, different measurements, different standards.

The map

Come back to this slide whenever you lose the thread



Regulation is the hinge between making something and selling it. Everything in this course hangs off one of these five boxes.

PART 1

The European framework

*Why a machine built in Perugia can be sold in
Hamburg without being tested twice*



The problem, before the solution

Europe, early 1980s

Every country wrote its own technical rules — for safety, for materials, for testing. Each rule was defensible on its own. Together they made a single European market impossible.

WHAT THAT COST

The same pressure vessel needed separate approval in France, Germany and Italy: three test campaigns, three sets of fees, three timelines — for one product. A small manufacturer simply did not export.

A national rule does not have to be protectionist to act as a trade barrier. It only has to be *different*.

The European Single Market

- Established **1 January 1993**
- Also called the Internal Market or the Common Market
- 27 EU member states, plus non-EU states through the EEA

The part that concerns engineers

Free movement of goods is achieved through *common rules and standards* that all participating states are legally committed to follow. That is where this course starts.

It guarantees the **four freedoms** — free movement of:

goods

capital

services

people

Three kinds of barrier

All three had to go

Physical

Border checkpoints, customs queues, paperwork at the frontier.

Technical

National rules, standards and approvals that differ from country to country.

Fiscal

Value-added tax and excise duties applied differently across states.

Ours is the middle one

Physical barriers were removed politically, fiscal barriers fiscally. The **technical** barrier is the engineering problem — and the only one that could not be solved by a signature alone.

The information procedure

One idea, three directives, forty years

Rather than forbidding national rules, Europe made them **transparent**: a state must announce a draft technical regulation before adopting it.

1

83/189/EEC

1983 — the original directive

2

98/34/EC

1998 — codified, extended to information society services

3

(EU) 2015/1535

in force since 7 October 2015 — the one that applies today

Why it still matters

The directive has been replaced twice, but the mechanism has not changed since 1983. Transparency, not prohibition, is what stops national rules fragmenting the market.

How the procedure works

1

Notification

The member state sends the draft technical regulation to the Commission and to the other member states.

2

Standstill

Three months in which it may not be adopted. The Commission and other states may comment.

3

Decision

In light of the comments, the state adopts, amends or drops the rule.

ONE EXCEPTION

An urgency procedure allows immediate adoption in serious and unforeseeable circumstances — a public health or safety emergency. The Commission decides whether the urgency is justified; if it accepts, the three-month standstill does not apply.

The New Approach, 1985

Council Resolution 85/C 136/01 — the idea that made it work

Harmonising every technical detail of every product in Europe by law was impossible. So the work was split in two.

Legislative harmonisation

Directives set only the *essential requirements* — mainly safety and health — that a product must meet. The law says *what*, never *how*. Meet them, and the product circulates freely.

Regulatory harmonisation

Standards bodies — CEN, CENELEC, ETSI — write the harmonised technical standards that say *how* to meet them. Voluntary, not law.

The consequence

Follow a harmonised standard and you are *presumed* to conform. You may prove conformity another way — but then the burden of proof is yours.

Mutual recognition

The principle

If a product is lawfully manufactured and marketed in one member state, there is no reason it should not be sold throughout the rest.

Member states may still issue their own mandatory technical rules where safety and health require it — but only if those rules comply with Community law and continue to guarantee free movement of goods. Policing that is exactly what the notification procedure does.

IN PRACTICE

Your machine is assessed once, in Italy, against a harmonised standard. It is then sold in Germany, Spain and Poland with no second assessment. That one sentence is worth more to a manufacturer than the whole rest of this lecture.

What it all produces: the CE marking

Regulation (EC) No 765/2008

- The manufacturer's **own declaration** that the product meets all applicable essential requirements
- It is *not* a quality mark, and *not* a certificate issued by an authority
- For higher-risk products a **notified body** must be involved before it may be affixed
- Affixing it without grounds is an offence across the member states

Regulation 765/2008 and Decision 768/2008/EC together form the *New Legislative Framework*, which also underpins national accreditation — and therefore ACCREDIA, which we meet next time.

LOOK AROUND THE ROOM

Every laptop, phone charger and wall socket here carries it. It is the most widely reproduced engineering symbol in Europe, and almost nobody can say what it legally means.

Where we got to

- A single market needs common rules — otherwise national technical rules fragment it
- Transparency, not prohibition: states notify drafts and wait three months
- The New Approach splits the work — law sets essential requirements, standards bodies say how to meet them
- Conform to a harmonised standard and you are presumed to comply
- The CE marking is the manufacturer's declaration that all of this was done

Next time

Who actually writes these standards — ISO, IEC, CEN, CENELEC, UNI, CEI — and how quality, reliability, certification and liability fit together. Start by going back to **the map**.

Questions?

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